# THE UNIFIED TRIAL EXAMINATION (2026)
# CHONG HWA INDEPENDENT HIGH SCHOOL K.L.

*Senior Middle Level*

## CHEMISTRY
**(SE11)**

## PAPER 1 MULTIPLE-CHOICE QUESTION

---

**Date: 18 September 2026**
**Time: 11:55 am – 12:55 pm**
**(60 MINUTES)**

---

### INSTRUCTIONS TO CANDIDTES

1. This subject comprises two papers:
Paper 1: Multiple-choice questions (30%)
Paper 2: Subjective questions (70%)

2. Complete Paper 1 within the 1 hour allocated. After 15-minute interval, proceed to complete Paper 2 within 1 hour 30 minutes allocated.

3. Attempt **all** the 30 questions in Paper 1. For each question, select the **correct** or the **best** answer and blacken the circle corresponding to the same letter of the alphabet as the answer you have selected. Mark the answers with a **5B** pencil on answer sheet 'O' (computer card).

4. The prescribed electronic calculators may be used.

5. The following data may be used in your calculations:
   * **Universal Gas Constant, R = 8.314 Pa $m^3$ $K^{-1}$ $mol^{-1}$**
   * **Faraday Constant, F = 96500 C $mol^{-1}$**

6. The Periodic Table of Elements sheet (not included in this copy).
7. Paper 1 consists of 7 printed pages.

**Do Not Turn Over This Page Until You Are Told To Do So**

Name: ( ) Class: 6 S/T ( )

*Prepared by and proof-read by: Tan Poh Poh; Chuah Meng Cheu; Gan Suk Ling; Tang Sin Yee*

1. The mass of one Carbon-12 atom is w g. What is the mass of one mole of Sodium-23 atoms?

A. $6.02 \times 10^{23} \times$ w g

B. $23/12 \times$ w g

C. $23 \times 6.02 \times 10^{23} \times$ w g

D. $23/12 \times 6.02 \times 10^{23} \times$ w g

2. The chemical formula of compound A is XY whereby the percentage of mass of X and Y in compound A are 77.78 % and 22.22 % respectively. Another compound B shows the chemical formula of $X_2Y_3$.

Which of the following is the **correct** percentage composition of X and Y in compound B?

| | **X** | **Y** |
|---|---|---|
| A. | 60 % | 40 % |
| B. | 70 % | 30 % |
| C. | 80 % | 20 % |
| D. | 85 % | 15 % |

3. Chromium forms an ion $Cr^{3+}$. What is the oxidation state and the valence electronic configuration of chromium in the ion $Cr^{3+}$?

| | Oxidation state | Valence electronic configuration |
|---|---|---|
| A. | +3 | $3d^3$ |
| B. | +3 | $3d^2\ 4s^1$ |
| C. | +3 | $3d^1\ 4s^2$ |
| D. | -3 | $3d^3$ |

4. Which of the following observations is **not** due to hydrogen bonding?

A. Ice is less dense than water.

B. Amino acids are water soluble.

C. The melting point of HBr is higher than HCl.

D. Dimerisation of ethanoic acid in organic solvent.

5. A gaseous mixture contains 44 g carbon dioxide and 64 g sulphur dioxide at 298 K and 101 kPa. Which of the following statement **best** describes the gaseous mixture?

A. The mixture contains more number of sulphur dioxide molecules.

B. The volume of carbon dioxide and sulphur dioxide gas is the same.

C. The average speed of carbon dioxide and sulphur dioxide molecules is the same.

D. The partial pressure of sulphur dioxide gas is higher than that of carbon dioxide gas.

6. A solution containing silver ion, $Ag^+$, was treated with an excess of chloride ions, $Cl^-$. 0.1234 g of AgCl precipitate was recovered. Assuming the percentage yield to be 98.7 %, how many grams of silver ions were present in the solution?

A. $1.409 \times 10^{-2}$ g B. $2.407 \times 10^{-2}$ g C. $3.401 \times 10^{-2}$ g D. $9.409 \times 10^{-2}$ g

7. The concentration of propanoic acid at 25°C is 0.05 mol $L^{-1}$ in diethyl ether and 0.20 mol $L^{-1}$ in water.

If 10 g of propanoic acid is dissolved in 100 mL of diethyl ether at 25°C, how many grams of propanoic acid can be extracted by adding 100 mL of water in a single extraction?

A. 2.0 g B. 5.0 g C. 8.0 g D. 9.5 g

8. Equation below shows an equilibrium system consisting of P, Q, R, and S.

$$P + Q \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} R + S \text{ ; } \Delta H \text{ positive}$$

$k_1$ represents rate constant for forward reaction, whereas $k_{-1}$represents the rate constant for reverse reaction. Which pair below about the effect of increasing temperature on the rate constant is **correct**?

| | $k_1$ | $k_{-1}$ |
|---|---|---|
| A | Increase | Increase |
| B | Increase | Decrease |
| C | Increase | No change |
| D | Decrease | Increase |

9. The molecule $N_2O_4$ is a dimer of the molecule $NO_2$. The enthalpy of formation of $N_2O_4$ and $NO_2$ are +9.67 kJ $mol^{-1}$ and +33.86 kJ $mol^{-1}$ respectively. Calculate the standard enthalpy of reaction, in kJ $mol^{-1}$, for the formation of $N_2O_4$ from $NO_2$.

A. +24.19 kJ $mol^{-1}$ B. +43.53 kJ $mol^{-1}$ C. -24.19 kJ $mol^{-1}$ D. -58.05 kJ $mol^{-1}$

10. Nitrogen dioxide decomposes on heat according to equation

$$2NO_2\,(g) \rightleftharpoons 2NO\,(g) + O_2\,(g)$$

When 4 moles of nitrogen dioxide was placed in a container of 1 L and was heated to a certain constant temperature, the equilibrium mixture contained 0.8 mol of oxygen. What is the numerical value for the equilibrium constant, Kc at the temperature of the experiment?

A. $\frac{0.8^2 \times 0.8}{4^2}$ B. $\frac{1.6 \times 0.8}{2.4^2}$ C. $\frac{1.6^2 \times 0.8}{4^2}$ D. $\frac{1.6^2 \times 0.8}{2.4^2}$

11. Consider the following equilibrium.

$$H_2S\,(g) + 2H_2O\,(l) \rightleftharpoons 2H_3O^+\,(aq) + S^{2-}\,(aq)$$

The concentration of $S^{2-}$ (aq) will decrease if

A. the concentration of $H_3O^+$ (aq) is increased.

B. $Na_2CO_3$ (s) is added.

C. $NH_4Cl$ (s) is added.

D. the pressure of the system is increased.

12. The pH value for a solution containing 0.245 mol HF in 500 mL of water is 1.88. Calculate the acid ionization constant of HF.

A. $1.64 \times 10^{-4}$ B. $3.55 \times 10^{-4}$ C. $8.83 \times 10^{-4}$ D. $9.64 \times 10^{-4}$

13. Sulphur dioxide dissolves in water to form a weak dibasic acid, $H_2SO_3$, which forms the following equilibria:

$$H_2SO_3 \rightleftharpoons H^+ + HSO_3^- \qquad K_{a1} = 1.5 \times 10^{-2}$$
$$HSO_3^- \rightleftharpoons H^+ + SO_3^{2-} \qquad K_{a2} = 1.0 \times 10^{-7}$$

Which of the following species are arranged in order of increasing concentration at equilibrium?

A. $[H_2SO_3]$, $[HSO_3^-]$, $[SO_3^{2-}]$

B. $[HSO_3^-]$, $[SO_3^{2-}]$, $[H_2SO_3]$

C. $[SO_3^{2-}]$, $[HSO_3^-]$, $[H_2SO_3]$

D. $[H_2SO_3]$, $[SO_3^{2-}]$, $[HSO_3^-]$

14. During a nuclear fission event, a uranium-235 nucleus absorbs a neutron and splits into barium-141 and krypton-92. How many neutrons are typically released in this specific reaction to conserve mass number?

A. 1 B. 2 C. 3 D. 4

15. Standard electrode potentials for copper and silver are given below.

$Ag^{+}(aq)/ Ag(s)$ $E° = +0.80$ V

$Cu^{2+}(aq)/ Cu(s)$ $E° = +0.34$ V

These metals are joined in a cell that could be used to obtain pure copper by electrolysis. When a current of 40.00 A is passed for 26.80 minutes, it is found that the anode decreases by 26.47 g. What is the percentage for composition of copper in the copper/ silver alloy?

A. 50% B. 60% C. 70% D. 80%

16. Sodium peroxide ($Na_2O_2$) reacts with water to produce $O_2$. This reaction is used in breathing masks, submarines, and spacecraft to supply oxygen. Which statement is correct?

A. This is a redox reaction. $Na_2O_2$ acts as both the oxidizing agent and the reducing agent.

B. This is a redox reaction. $Na_2O_2$ is the oxidizing agent and water is the reducing agent.

C. This is a redox reaction. $Na_2O_2$ is the reducing agent and water is the oxidizing agent.

D. This is a displacement reaction because $O_2$ is produced.

17. Equal amounts of aluminum powder react separately with excess hydrochloric acid and excess concentrated sodium hydroxide solution. Under the same conditions, what is the volume ratio of the gases produced?

A. 1:1 B. 1:2 C. 1:3 D. 3:2

18. Which experimental fact proves that silicic acid is weaker than carbonic acid?

A. $CO_2$ dissolves in water to form carbonic acid, while $SiO_2$ is insoluble.

B. Passing $CO_2$ into a soluble silicate solution produces a silicic acid precipitate.

C. At high temperature $SiO_2$ reacts with carbonates to produce $CO_2$.

D. HCl releases gas from soluble carbonates but forms precipitate with soluble silicates.

19. Coloured cloth soaked in bleaching powder solution shows better bleaching after being left in air because:

A. The bleaching solution is oxidized.

B. Oxygen oxidizes the colored cloth.

C. Bleaching powder reacts with $CO_2$ in air to form more HClO.

D. Water evaporates, increasing concentration.

20. Which solution can be used to test whether a gas jar is filled with $SO_2$?

A. $BaCl_2$ solution
C. Starch-potassium iodide solution
B. Acidified $KMnO_4$ solution
D. Limewater

21. Which statement about concentrated sulphuric acid is **correct**?

A. It carbonizes sugar because of its drying property.
B. It rapidly reacts with copper at room temperature to release hydrogen.
C. It can dry ammonia and hydrogen.
D. It passivates iron and aluminum by forming an oxide layer at room temperature.

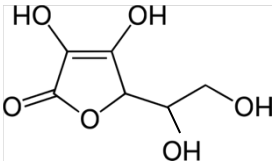
22. Which single reagent can distinguish NaCl, $AlCl_3$, $FeCl_2$, $FeCl_3$ and $MgCl_2$ in one step?

A. KSCN
B. $BaCl_2$
C. NaOH
D. HCl

23. After complete reaction, which mixture gives a red colour with potassium thiocyanate solution?

A. Iron powder with hydrochloric acid
B. Iron(II) sulphide with dilute sulphuric acid
C. Red-hot iron wire with chlorine gas
D. Iron(III) chloride solution with excess iron powder

24. Vitamin C has the structural formula shown below.




How many chiral carbons are present in one molecule of vitamin C?

A. 0
B. 1
C. 2
D. 3

25. The structural formulae of three organic compounds are given below.

| $CH_3CH_2CH_2OH$ | $CH_3CH_2Cl$ | $CH_3COOH$ |
|---|---|---|
| X | Y | Z |

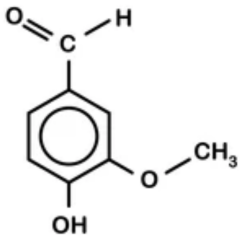
Which sequence shows these compounds in order of increasing boiling point?

A. X < Y < Z
B. Y < X < Z
C. Y < Z < X
D. Z < X < Y

26. Which alkene will form two different carboxylic acids when heated with acidified potassium manganate(VII) solution?

A. $CH_3CH_2CH=CH_2$
B. $CH_3CH=CHCH_2CH_3$
C. $(CH_3)_2C=CHCH_3$
D. $CH_3CH_2CH=CHCH_2CH_3$

27. Vanillin has the structural formula shown below:




It is often used as a flavoring agent in food, beverages, and pharmaceuticals. Which of the following is **not** a reaction of vanillin?

A. It forms an orange precipitate when reacted with 2,4-dinitrophenylhydrazine.
B. It gives a white precipitate when treated with excess aqueous bromine.
C. It decolorizes hot acidified potassium manganate(VII) solution.
D. It forms a reddish-brown precipitate when warmed with Fehling's solution.

28. One of the esters present in pineapples is compound P, $CH_3CH_2CH_2COOCH_3$. When compound P is heated under reflux with dilute aqueous sodium hydroxide, which organic products are formed?

I. $CH_3CH_2CH_2CH_2OH$
II. $CH_3CH_2CH_2COONa$
III. $CH_3OH$
IV. $CH_3ONa$

A. I and II
B. II and III
C. III and IV
D. I and IV

29. Which of the following compounds undergoes self-condensation to form a polymer?

A. $CH_3CH=CH_2$
B. $HOCH_2CH_2OH$
C. $H_2NCH_2CH_2NH_2$
D. $HOCH_2CH_2COOH$

30. A polymer has the following repeating unit:

$$—[-CH_2\text{-}CHCl—CH_2\text{-}CH=CH\text{-}CH_2-]_n—$$

Which pair of monomers can be polymerized to produce this polymer?

A. $CH_2=CHCl$ and $CH_2=CHCH=CH_2$
B. $CH_2=CClCH=CH_2$ and $CH_2=CH_2$
C. $CH_2=CHCl$ and $CH_2=CH_2$
D. $CH_3CCl=CH_2$ and $CH_3CH=CH_2$